

Supplementary Materials

When does sampling uncertainty matter in matrix population models? Evidence from published projection matrices

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46 **Supplementary Methods**

47 *S1.1 Sampling-distribution workflow used for each MPM*

48 ~~We reconstructed transition-rate uncertainty by direct conjugate posterior sampling, using 2,000~~
49 ~~draws per MPM for the illustrative example and 1,000 for Analysis 1. For each origin-stage~~
50 ~~column of U , allowed living transitions and death were drawn jointly from $\text{Dirichlet}(k_1 +$~~
51 ~~$1, \dots, k_m + 1)$, corresponding to a multinomial likelihood and a $\text{Dirichlet}(1, \dots, 1)$ prior. Counts~~
52 ~~k_m were reconstructed as transition rates multiplied by stage sample size, after rescaling any U~~
53 ~~column whose sum exceeded 1. These reconstructed counts may be fractional for rounded or~~
54 ~~averaged matrix entries. Each allowed reproductive rate was drawn from $\text{Gamma}(1 + k, \text{rate} =$~~
55 ~~$n)$, where $k = nF_{ij}$, corresponding to a Poisson likelihood and an improper flat prior on the~~
56 ~~non-negative rate scale. These are posterior parameter draws, not bootstrap resamples of~~
57 ~~observations. We rebuilt each MPM and recalculated derived quantities for every draw.~~

58 ~~Allowed-transition masks default to positive entries in the focal matrix; where masks are~~
59 ~~constructed across matrices, an entry is allowed if positive in any contributing matrix. Other~~
60 ~~entries remain zero. Death is included even when its reconstructed count is zero, provided the~~
61 ~~column has an allowed living transition. This does not establish that excluded zero-valued~~
62 ~~transitions are biologically impossible. A missing sample size leaves its column unchanged; a~~
63 ~~sample size of zero produces a missing simulated column. Neither case supplies a posterior~~
64 ~~uncertainty estimate for that column. Figure 1C displays normalised marginal binomial or~~

Deleted: For each focal matrix population model (MPM), we reconstructed transition-level sampling distributions from reported transition counts (or equivalent sample-size information). We then generated 2,000 parametric bootstrap draws per MPM by sampling each transition from its likelihood (multinomial for survival/growth/retrogression transitions and Poisson for recruitment transitions), rebuilding the full MPM for each draw, and recalculating each derived demographic quantity from that draw. We treated structural zeros (biologically impossible transitions) as fixed zeros in all draws, and treated sampling zeros (possible but unobserved transitions) as uncertain parameters under the same likelihood models used for non-zero transitions. When a recovered stage-specific sample size was zero, we retained the published point-estimate transition column unchanged in all bootstrap draws rather than treating it as structurally fixed at zero; this preserves the published matrix while acknowledging that no additional sampling variance can be propagated for that column.⁴ This produced a sampling distribution for each derived quantity used downstream (e.g. life expectancy, shape, log population growth rate). Point estimates in the manuscript

86 Poisson likelihood curves under flat priors over the plotted range; these are distinct from the
87 joint Dirichlet draws used for derived quantities.

88 Point estimates correspond to values derived directly from the reported matrix entries;
89 uncertainty-aware summaries describe the reconstructed posterior distributions.

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90 *S1.2 Age-from-stage derivation, truncation, and shape metric*

91 Life expectancy and shape were derived from survivorship trajectories obtained using age-from-
92 stage methods, starting from the first reproductive stage. Shape was calculated using the pace-
93 shape framework metric (Baudisch and Stott 2019). Because stage-based survivorship can
94 converge to an artefactual mortality plateau, shape was calculated on truncated trajectories up to
95 quasi-stable convergence (distance threshold 1%).

96 *S1.3 Pooled and mean MPM handling*

97 Analysis 1 used one population-level MPM per population. Where a pooled population MPM
98 was available, it was used; otherwise, a mean population MPM was used; if neither was available,
99 a population-level matrix was constructed as the element-wise mean of the available annual
100 individual MPMs. When COMPADRE included both site-specific population rows and
101 aggregate rows that combined multiple populations, only the site-specific population rows were
102 retained, so that each population contributed a single analysis unit. For uncertainty propagation
103 in pooled or mean matrices, stage-specific sample sizes were pooled across years before posterior
104 sampling. This captures within-year sampling variation but does not fully represent between-year
105 process variation, so the resulting uncertainty should be interpreted conservatively.

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106 *S1.4 Bayesian model priors and fitting settings*

107 Stan models and priors are defined in `models/*.stan`. For the three main analyses:

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- Analysis 1 models use normal priors for location/slope terms and half-normal priors for positive scale terms (see `models/regress_hier.stan` and `models/regress_hier_error.stan`).

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- Analysis 2 uses Gaussian-process recruitment models with normal priors on GP hyperparameters and coefficients (`models/movbeta_gprc.stan` and `models/movbeta_gprc_err.stan`), plus spring-temperature models (`models/regress.stan` and `models/regress_log_err.stan`).

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- Analysis 3 fits its model with year-specific process variation using `models/poisson_climate_latent.stan`: the intercept and climate coefficient have Normal(0,2) and Normal(0,1) priors, respectively; the process standard deviation has an Exponential(1) prior, and the standardised year effects have Normal(0,1) priors. Its Gaussian and Poisson point-estimate models use `stats::glm()` and have no Bayesian priors.

- The multisite *Astragalus scaphoides* robustness analysis uses the same Gaussian-process recruitment model pair as Analysis 2 (`models/movbeta_gprc.stan` and `models/movbeta_gprc_err.stan`) together with a simpler spring-temperature regression pair using weakly informative normal priors on regression coefficients and scale terms (`models/regress.stan` and `models/regress_log_err.stan`).

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Fitting settings used in scripts:

- Analysis 1 (`scripts/S07_case_study_1_analysis.R`): `chains = 2`, `iter = 4000`, `warmup = 2000-3000`, `seed = 5654`.
- Analysis 2 (`scripts/S12_case_study_2.R`): `iter = 5000` for key models, `seed = 5654`, with additional control settings for GP fits.

- Analysis 3 (`scripts/S16_analysis_3_simulation.R`): `chains = 4, iter = 3000, warmup = 1500`, and a separate reproducible seed for each simulated time series.
- Multisite *Astragalus scaphoides* robustness analysis (`scripts/S14_case_study_3_astragalus.R`): `iter = 5000, seed = 5654`, using the same GP control settings as Analysis 2 for recruitment models and matched regression settings for the simplified spring-temperature models.

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S1.5 Comparison of point-estimate and posterior-mean variance

Deleted: S1.5 Variance-component summary (sampling-to-other ratio)

For each parameter, we compared variance across population-level MPM entries using:

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- the value derived from each matrix point estimate, θ_p ;
- the mean of its reconstructed posterior distribution, θ_{p-} .

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For population-level entries p , the reported ratio is Eq. (1):

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$$\text{ratio} = \frac{\text{Var}_p(\theta_p)}{\text{Var}_p(\theta_{p-})} \quad (1)$$

Deleted: ratio = $\frac{\text{Var}_{\text{sampling}}}{\text{Var}_{\text{residual}}} (1)$
Ratios above 1 indicate that sampling uncertainty exceeds residual biological variation for that parameter.

Ratios above 1 indicate that point estimates are more dispersed than posterior means. We obtained 95% intervals by resampling population-level entries with replacement 2,000 times, keeping each point estimate paired with its posterior mean. This descriptive ratio does not say how much of that spread is due to sampling noise rather than genuine biological differences.

S1.6 Simplified spring-temperature model (Analysis 2)

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In addition to the monthly-lagged GP model, we fit a simplified spring summary model with mean spring temperature deviation as the predictor, under both point-estimate and uncertainty-aware formulations. The contrasts reported in the manuscript use posterior medians and 95% interval widths from these paired models.

174 For the stage-specific survival panel in [Analysis 2](#) (Fig. 5C), we derived annual stage-specific
 175 survival as the column sums of the U matrix for each year and stage, then compared two logit-
 176 scale models per stage: (i) a stable-stage distribution (SSD)-weighted quasibinomial point-
 177 estimate model, $\text{surv} \sim \text{tmp}$, where weights are the stable stage distribution from each annual U
 178 matrix; and (ii) a binomial count model, $\text{cbind}(n_{\text{surv}}, n_{\text{fail}}) \sim \text{tmp}$, where n_{surv} and n_{fail} are
 179 survived and non-survived counts implied by stage sample size and U -column survival
 180 probability. The quasibinomial model estimates dispersion, whereas the binomial model fixes it
 181 at 1; together with the different weights, this prevents attributing interval-width differences solely
 182 to sampling uncertainty. Intervals are coefficient estimates ± 1.96 standard errors (95% Wald
 183 confidence intervals). Stages failing the fitting criteria (at least five usable years, variation in
 184 temperature, and both survivors and deaths) are labelled as not estimable.

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185 *S1.7 Reporting recommendations for future MPM studies*

186 To support uncertainty propagation in secondary analyses, primary studies should report (at a
 187 minimum):

- 188 • transition-level sample sizes for each matrix element (or reconstructable transition count
 189 tables);
- 190 • clear identification of pooled/mean matrices and pooling rules;
- 191 • uncertainty estimates on model-appropriate scales (e.g. linear predictor where relevant);
- 192 • explicit projection interval, stage definitions, and treatment filters used to derive
 193 published matrices.

194 These items are usually sufficient for parametric uncertainty reconstruction in compiled
 195 databases.

198 *S1.8 Included versus excluded studies in the current target-study screen*

199 To assess whether the studies retained for uncertainty analysis differed systematically from those

200 screened but not retained, we compared the 23 post-2010 studies in the final dataset that still fall

201 within the current reproducible COMPADRE screen against the remaining 69 screened studies.

202 For each study, we summarized mean matrix dimension, mean stage-specific survival (the

203 average of *U*-column survival probabilities across matrices), and the proportions of stage-specific

204 survival estimates equal to 0, equal to 1, or at either boundary. Dominant life form was defined

205 as the modal **OrganismType** among a study's matrices. Three additional post-2010 studies in the

206 final dataset do not fall within the current screen under COMPADRE version 6.26.3.0 (created

207 Mar 11 2026) (*Paeonia officinalis*, *Tillandsia butzii*, *Lantana camara*), and five further studies were

208 pre-2010 additions assembled on the basis of temporal replication (at least four annual matrices);

209 these eight studies were excluded from this screen-based comparison. The requirement of at

210 least four annual matrices applied only to those five pre-2010 studies, which were added to

211 support the time-series analyses; it was not a selection criterion for the full 31-study dataset used

212 across all analyses.

213 *S1.9 Study and species identities in Analysis 1*

214 The Analysis 1 life expectancy-shape models used 68 population-level matrix entries representing

215 25 species. This is a different subset from the boundary/small-sample audit in section S1.11.

216 Study-level and species-level identities (including COMPADRE **SpeciesAuthor** labels) are

217 provided in Table S6.

218 *S1.10 Multisite robustness extension of Analysis 2 (Astragalus scaphoides)*

219 To evaluate whether the Analysis 2 conclusions generalised across populations, we repeated the

220 same paired modelling framework for all available *Astragalus scaphoides* sites with sufficient annual

221 coverage as a supplementary robustness analysis. For each site, we fit (i) a point-estimate model

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Deleted: used a subset of the full compiled dataset:

Deleted: 9 Multi-site

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228 and (ii) a sampling-uncertainty model for recruitment-temperature relationships, using both the
 229 monthly-lag Gaussian-process specification and the simplified spring-temperature specification.
 230 We then compared posterior medians and 95% intervals between model formulations within
 231 site. This design ~~compares model formulations within each site rather than isolating the~~
 232 contribution of uncertainty propagation.
 233 For stage-specific survival in ~~this supplementary robustness analysis~~, we used the same pair of
 234 formulations as in ~~Analysis 2~~: an SSD-weighted quasibinomial point-estimate model ($\text{surv} \sim$
 235 tmp , weights from the stable stage distribution of each annual U matrix) and a binomial count
 236 model ($\text{cbind}(n_{\text{surv}}, n_{\text{fail}}) \sim \text{tmp}$). ~~The linear predictor and logit link are shared, but weighting~~
 237 ~~and dispersion assumptions also differ, as described in section S1.6; the comparison therefore~~
 238 ~~does not isolate~~ sampling uncertainty.
 239 For sensitivity analyses intended to mimic database-only use (no access to true stage sample
 240 sizes), we re-fit stage-specific point-estimate models under two alternatives: (i) no weighting and
 241 (ii) SSD weighting derived from each annual U matrix.

242 *S1.11 Boundary survival estimates and small stage-specific sample sizes in Analysis 1*
 243 *To assess whether the conditions most likely to distort point estimates were concentrated in*
 244 *particular parts of the Analysis 1 dataset, we summarized a separate set of 68 population-level*
 245 *entries from 28 species with recovered stage counts by COMPADRE OrganismType (used here*
 246 *as the available life-form grouping) and by broad matrix-dimension class (3-5 versus 6+ stages).*
 247 *For each population-level matrix, we recorded whether any stage-specific survival estimate lay*
 248 *exactly on the boundary (0 or 1), whether any stage had $N < 20$ or $N < 50$, and the minimum*
 249 *stage-specific sample size. This audit used unscaled U matrices, with exactly 100% survival in*
 250 *22/68 entries, rather than the rescaled matrices and model subset described in the main*

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Deleted: response scale aligned across formulations, differing only...

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Discussion. These summaries are descriptive and intended to identify where potentially fragile point estimates are concentrated in the analysed subset.

S1.12 Displacement magnitude versus boundary survival structure

To examine factors associated with point-estimate displacement in Analysis 1, we compared each population-level point estimate to the median of its sampling distribution for mature life expectancy and shape. We defined displacement magnitude as the absolute difference between the point estimate and the sampling-distribution median; for life expectancy, this was calculated on the $\log_{10}$ scale to reflect multiplicative discrepancies. We then related displacement magnitude to two predictors derived from the same population-level matrix: (i) the number of stage-specific survival estimates at the boundary (0 or 1), and (ii) the proportion of observed individuals belonging to boundary-survival stages, calculated from the recovered stage-specific sample sizes. We summarized these relationships with simple linear regressions and visualized them with scatter plots and fitted lines.

S1.13 Sensitivity to the quasi-stable-distribution truncation threshold

To assess whether the shape metric was sensitive to the truncation rule used to avoid artefactual mortality plateaus, we repeated the Analysis 1 shape calculation using quasi-stable-distribution convergence thresholds of 0.5% and 5%, alongside the 1% threshold used in the main analysis. For each population-level MPM that remained estimable under all three thresholds, we recalculated the point estimate of shape and the posterior uncertainty summary using the threshold-specific truncation age q . We then compared the resulting shape estimates to the 1% baseline using pairwise correlations, absolute differences, sign changes around zero, and shifts in q .

284 **Supplementary Results**

285 *S2.1 Sensitivity to point-estimate weighting choice*

286 Across estimable stages in [Analysis 2](#), switching from no weighting to SSD weighting changed
287 temperature-coefficient medians by at most 0.142 on the logit scale, with interval widths
288 changing by -0.045 to 0.614 (Table S9).

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289 Across estimable site-stage combinations in [the *Astragalus robustus* analysis](#), the corresponding
290 maximum absolute median shift was 0.912, and interval-width differences ranged from -0.512 to
291 1.526 (Table S10). These sensitivity checks show how weighting assumptions alter point-estimate
292 uncertainty when true stage sample sizes are unavailable.

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293 *S2.2 Where do boundary estimates and small stage counts occur?*

294 [Among the 68 Analysis 1 population-level entries with recovered stage counts, 42.6% contained](#)
295 [at least one boundary survival estimate and 30.9% contained at least one stage estimated from](#)
296 [fewer than 20 individuals. These conditions tended to co-occur: the population-level proportion](#)
297 [of boundary survival estimates was positively correlated with the proportion of stages having](#)
298 [N < 20 \(r = 0.35\).](#)

299 [By life form, herbaceous perennials made up most of the analysed subset and showed the highest](#)
300 [frequency among the well-represented categories, with 45.3% of entries containing at least one](#)
301 [boundary survival estimate and 34.0% containing at least one stage with N < 20. Succulents](#)
302 [showed lower corresponding values \(22.2% and 22.2%, respectively\). Woody life forms were too](#)
303 [sparsely represented in the final subset for strong conclusions. By matrix dimension, both](#)
304 [features were more common in 6+-stage matrices than in 3-5-stage matrices \(61.1% versus](#)
305 [36.0% for any boundary estimate; 44.4% versus 26.0% for any stage with N < 20; Table S12\).](#)

Boundary survival estimates can arise from finite sampling of discrete stage classes. Finer stage classes do not necessarily reduce these estimates and may leave fewer observations per class. Information sharing in fitted vital-rate models, such as those used to construct integral projection models, may reduce boundary estimates through smoothing rather than finer discretisation alone.

S2.3 Does displacement increase with boundary survival structure?

Displacement magnitude increased with both measures of boundary-survival structure (Figure S3; Table S13). For shape, the absolute difference between the point estimate and the sampling-distribution median increased with the number of boundary survival stages (slope = 0.010, 95% CI [0.005, 0.016]) and with the proportion of individuals in boundary-survival stages (slope = 0.093, 95% CI [0.043, 0.143]). For mature life expectancy, the same pattern was evident on the $\log_{10}$ scale (number of boundary stages: slope = 0.062, 95% CI [0.029, 0.095]; proportion of individuals in boundary-survival stages: slope = 0.672, 95% CI [0.370, 0.974]). The fitted models explained a modest but non-trivial share of the among-matrix variation in displacement magnitude ($R^2 = 0.176 - 0.242$). These associations are consistent with the proposed mechanism, but do not establish causality.

S2.4 How sensitive are shape estimates to the truncation threshold?

In the 65 population-level entries for which shape was estimable under all three thresholds, the shape metric was highly stable to the quasi-stable-distribution truncation rule (Table S14; Figure S4). Relative to the 1% threshold used in the main analysis, the 0.5% threshold changed point estimates by a median absolute amount of 0.002 and sampling-distribution medians by 0.001. The more permissive 5% threshold produced somewhat larger but still modest shifts (median absolute changes 0.004 and 0.003, respectively), with only 2 point estimates changing sign relative to the 1% baseline. Those sign changes occurred only for populations whose baseline

shape estimates were already very close to zero. These results indicate that shape estimates were generally stable across the thresholds tested. We did not refit the downstream regressions, so their sensitivity to threshold choice remains untested.

*S2.5 Multisite robustness analysis in *Astragalus scaphoides**

For recruitment, applying the Analysis 2 framework to three *Astragalus scaphoides* populations yielded the same overall pattern: point-estimate and sampling-uncertainty models were closely aligned within site, while differences among sites were more pronounced than differences between model formulations (Figure S5). Across sites, we had 9-11 years of annual demographic records, with 8-10 years retained for model fitting after filtering (0-3 years excluded; Table S7). For spring-temperature models, posterior medians differed only slightly between formulations (maximum absolute within-site difference: 0.026 on the linear predictor scale), and 95% interval widths changed by only -5.3% to 4.7% across sites (Table S8). Monthly-lag coefficient profiles were likewise highly concordant between formulations, with within-site correlations of posterior medians ranging from 0.992 to 0.997 and a maximum lag-specific median difference of 0.042 (Figure S5). These results support the main-text conclusion from Analysis 2 that climate-effect inference can be robust to transition-rate sampling uncertainty in this empirical setting.

Supplementary Tables

Table S1. Single-MPM derived-parameter summary for the focal *Agrimonia eupatoria* example. Reported values are point estimates and 95% credible intervals from reconstructed posterior distributions, showing substantial uncertainty for several derived demographic quantities.

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Parameter	Median	Lower 95%	Upper 95%	Point estimate
Pop. growth rate (λ)	0.943	0.826	1.010	0.956
Damping ratio (ρ)	1.440	1.110	2.200	1.310

Parameter	Median	Lower 95%	Upper 95%	Point estimate
Reproductive value (vegetative stage)	2.800	0.803	15.000	
Elasticity	0.345	0.145	0.558	
Generation time (T)	25.900	15.300	58.100	31.500
Life expectancy (l_0)	6.650	3.600	17.900	7.840

354

355 Table S2. Mean-MPM derived-parameter summary for the focal *Agrimonia eupatoria* example
356 (population B). Point estimates are compared with sampling-distribution summaries to show
357 how pooling/averaging changes central tendency and uncertainty relative to the single-MPM
358 case.

Parameter	Median	Lower 95%	Upper 95%	Point estimate
Life expectancy (l_0)	1.920	1.69	2.16	1.930
Pop. growth rate (λ)	0.929	0.82	1.03	0.958
Damping ratio (ρ)	1.430	1.25	1.68	1.370

359

360 Table S3. Analysis 1 posterior summary for the slope parameter (β) in the life expectancy-shape
361 model. The uncertainty-aware model shows greater posterior support for $\beta > 0$ than the point-
362 estimate model.

Model	Median beta	Lower 95%	Upper 95%	P(beta > 0)	
Model using point estimates	-0.016	-0.052	0.023	0.206	Deleted: 017
Model with sampling uncertainty	0.011	-0.017	0.038	0.770	Deleted: 053
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					Deleted: 146

368 Table S4. Analysis 2 Gaussian-process regression coefficients (monthly lags) for recruitment
369 responses to temperature anomalies. Lag-specific posterior summaries are similar between point-
370 estimate and uncertainty-aware model formulations.

Lag	Model	Median beta	Lower 95%	Upper 95%	lag_err
1	Point estimates	-0.251	-0.659	0.154	0.82
2	Point estimates	-0.135	-0.472	0.223	1.82
3	Point estimates	-0.030	-0.358	0.282	2.82
4	Point estimates	0.078	-0.261	0.396	3.82
5	Point estimates	0.155	-0.196	0.474	4.82
6	Point estimates	0.207	-0.156	0.507	5.82
7	Point estimates	0.272	-0.040	0.563	6.82
8	Point estimates	0.331	0.055	0.608	7.82
9	Point estimates	0.320	0.054	0.538	8.82
10	Point estimates	0.245	-0.068	0.482	9.82
11	Point estimates	0.200	-0.154	0.460	10.80
12	Point estimates	0.145	-0.151	0.416	11.80
13	Point estimates	0.111	-0.154	0.407	12.80
14	Point estimates	0.049	-0.195	0.411	13.80
15	Point estimates	-0.003	-0.251	0.421	14.80
16	Point estimates	-0.091	-0.333	0.197	15.80
17	Point estimates	-0.182	-0.464	0.084	16.80
18	Point estimates	-0.203	-0.543	0.046	17.80
19	Point estimates	-0.203	-0.494	0.044	18.80
20	Point estimates	-0.159	-0.393	0.062	19.80
21	Point estimates	-0.082	-0.306	0.181	20.80
22	Point estimates	-0.009	-0.278	0.291	21.80
23	Point estimates	0.018	-0.317	0.361	22.80
24	Point estimates	-0.002	-0.451	0.414	23.80
1	Sampling uncertainty	-0.274	-0.735	0.189	1.18

Lag	Model	Median beta	Lower 95%	Upper 95%	lag_err
2	Sampling uncertainty	-0.163	-0.551	0.236	2.18
3	Sampling uncertainty	-0.035	-0.402	0.329	3.18
4	Sampling uncertainty	0.079	-0.265	0.464	4.18
5	Sampling uncertainty	0.162	-0.197	0.517	5.18
6	Sampling uncertainty	0.225	-0.136	0.554	6.18
7	Sampling uncertainty	0.295	-0.038	0.615	7.18
8	Sampling uncertainty	0.348	0.061	0.631	8.18
9	Sampling uncertainty	0.342	0.078	0.571	9.18
10	Sampling uncertainty	0.277	-0.068	0.543	10.20
11	Sampling uncertainty	0.216	-0.161	0.497	11.20
12	Sampling uncertainty	0.155	-0.204	0.456	12.20
13	Sampling uncertainty	0.117	-0.168	0.401	13.20
14	Sampling uncertainty	0.053	-0.215	0.448	14.20
15	Sampling uncertainty	-0.004	-0.270	0.406	15.20
16	Sampling uncertainty	-0.100	-0.338	0.207	16.20
17	Sampling uncertainty	-0.182	-0.473	0.068	17.20
18	Sampling uncertainty	-0.226	-0.568	0.069	18.20
19	Sampling uncertainty	-0.226	-0.548	0.043	19.20
20	Sampling uncertainty	-0.193	-0.413	0.054	20.20
21	Sampling uncertainty	-0.112	-0.356	0.183	21.20
22	Sampling uncertainty	-0.043	-0.333	0.335	22.20
23	Sampling uncertainty	-0.020	-0.376	0.362	23.20
24	Sampling uncertainty	-0.013	-0.456	0.416	24.20

372 Table S5. Analysis 2 posterior summary for the spring-temperature coefficient. Point-estimate
373 and uncertainty-aware models yield similar medians and interval widths, indicating limited effect
374 of uncertainty propagation on this coefficient.

Model	Median	Lower 95%	Upper 95%
Point estimates	1.12	0.135	2.10
Sampling uncertainty	1.21	0.151	2.33

375

376 Table S6. Primary studies represented in the analysis subset, as identified in COMPADRE. For
 377 matrix-level metadata and source-link details beyond this summary, refer to the COMPADRE
 378 database record for each study.

Authors	Year	Journal	DOI	Species (COMPADRE label)
Andrello	2010			Dracocephalum_austriacum
Dangremond; Knight	2010	Ecology	10.1890/09-0418.1	Lupinus_tidestromii
Davison; Jacquelyn; Adriaens; Honnay; de Kroon; Tuljapurkar	2010	J Ecol	10.1111/j.1365-2745.2009.01611.x	Anthyllis_vulneraria
Fortini; Zarin	2010	Forest Ecol Manag	10.1016/j.foreco.2010.11.007	Callycophyllum_spruceanum; Carapa_guianensis; Licania_heteromorpha; Licaria_mahuba; Mora_paraensis; Platymiscium_filipes; Virola_surinamensis
Gaoue; Ticktin	2010	Conserv Biol	10.1111/j.1523-1739.2009.01345.x	Khaya_senegalensis
Jacquelyn; Brys; Jongejans	2010	Ecology	10.1890/08-2321.1	Orchis_purpurea
Picard; Mortier; Chagneau	2010	Ecol Model	10.1016/j.ecolmodel.2010.06.010	Dicorynia_guianensis
Sampaio; Scarlet	2010	J Trop Ecol	10.1017/S0266467409990599	Geonoma_schottiana
Sletvold; Øien; Moen	2010	Biol Conserv	10.1016/j.biocon.2009.12.017	Dactylorhiza_laponica
Bullock; White; Prudhomme; Tansey; Perea; Hooftman	2011	J Ecol	10.1111/j.1365-2745.2011.01910.x	Cirsium_acaule_2
Che-Castaldo; 2011		Ecosphere	10.1890/ES11-00263.1	Frasera_speciosa_2
Inouye				
Chen; Visser; Jongejans; van Breugel; Zuidema; Kassim; de Kroon	2011	J Ecol	10.1111/j.1365-2745.2011.01825.x	Shorea_leprosula
Csergő; Molnár; García	2011	Popul Ecol	10.1007/s10144-010-0249-y	Saponaria_bellidifolia
Dominguez	2011	J Nat Conserv	10.1016/j.jnc.2011.05.005	Vella_pseudocytisus_subsp._pau_2

Authors	Year	Journal	DOI	Species (COMPADRE label)
Lozano; Moreno Saiz; Schwartz Navarro; Galeano; Bernal	2011	Trop Conserv Sci	10.1177/194008291100400104	Lepidocaryum_tenua
Norghauer; Newbery Thorpe; Stanley; Kayne; Latham Andrello; Bizoux; Barbet-Massin; Gaudeul; Nicolé; Till-Bottraud	2011	Ecol Monogr	10.1890/10-2268.1	Microberlinia_bisulcata; Tetraberlinia_bifoliolata
	2011			Cypripedium_fasciculatum
	2012	Biol Conserv	10.1016/j.biocon.2011.12.012	Eryngium_alpinum
Ellis	2012	Ecology	10.1890/11-1052.1	Liatris_scariosa
Ellis; Williams; Lesica; Bell; Bierzychudek; Bowles; Crone; Doak; Ehrlén; Ellis-Adam; McEachern; Ganesan; Latham; Luijten; Kaye; Knight; Menges; Morris; den Nijs; Oostermeijer; Quintana-Ascencio; Shelly; Stanley; Thorpe; Ticktin; Valverde; Weekley	2012	Ecology	10.1890/11-1052.1	Cirsium_pitcheri_4; Phyllanthus_emblica_3
Groenen dijk; Eshete; Sterck; Zuidema; Bongers	2012	J Appl Ecol	10.1111/j.1365-2664.2011.02078.x	Boswellia_papyrifera
Scanga; Leopold	2012	Biol Conserv	10.1016/j.biocon.2012.01.061	Trollius_laxus
Ticktin; Ganesan	2012	J Appl Ecol	10.1111/j.1365-2664.2012.02156.x	Phyllanthus_emblica_2; Phyllanthus_indofischeri

Authors	Year	Journal	DOI	Species (COMPADRE label)
Parameha; Setty Gornish	2013			Pityopsis_aspera
Matlagas; Davis	2013	J Appl Ecol	10.1111/1365-2664.12057	Miscanthus_giganteus
Osunkoya; Perrett; Fernando; Clark; Raghu Vilellas; Ehrlén; Olesen; Braza; García	2013	Popul Ecol	10.1007/s10144-013-0364-7	Lantana_camara
	2013	Ecography	10.1111/j.1600-0587.2012.07425.x	Plantago_coronopus_2
Scanga	2014	Plant Ecol	10.1007/s11258-014-0344-9	Trollius_laxus_2
Baldauf; Correa; Ferreira; Santos	2015	Forest Ecol Manag	10.1016/j.foreco.2015.06.022	Himatanthus_drasticus
Gibert; Magda; Hazard	2015	PLOS ONE	10.1371/journal.pone.0139919	Festuca_eskia
Berry; Natalie L Cleavitt	2021	Popul Ecol	10.1002/1438-390X.12092	Platanthera_macrophylla_; Platanthera_orbiculata_

380 Table S7. [Multisite robustness-analysis](#) site coverage and spring-temperature effect summaries
381 for *Astragalus scaphoides*. Point-estimate and sampling-uncertainty models give similar spring-
382 temperature coefficients within each site, showing that between-site heterogeneity is larger than
383 within-site method differences.

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Site	Years (total)	Years (modelled)	Years excluded	Spring beta (point)	Spring beta (sampling)
Haynes Creek	11	8	3	-0.441	-0.424
Sheep Corral Gulch	11	10	1	0.244	0.258
McDevitt Creek	9	9	0	-0.238	-0.212

384

386 Table S8. [Multisite robustness-analysis](#) posterior summary for the spring-temperature coefficient
387 by site and model formulation. Across all sites, 95% intervals overlap zero under both
388 formulations, showing similar uncertainty conclusions with and without explicit sampling-
389 uncertainty propagation.

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Site	Model	Median	Lower 95%	Upper 95%
Haynes Creek	Point-estimate model	-0.441	-1.76	0.899
Haynes Creek	Sampling-uncertainty model	-0.424	-1.74	0.872
McDevitt Creek	Point-estimate model	-0.238	-1.45	1.110
McDevitt Creek	Sampling-uncertainty model	-0.212	-1.52	1.160
Sheep Corral Gulch	Point-estimate model	0.244	-1.69	2.100
Sheep Corral Gulch	Sampling-uncertainty model	0.258	-1.51	2.080

390

392 Table S9. Analysis 2 stage-specific survival-temperature coefficients under two point-estimate
393 weighting scenarios. “No weights” treats annual stage-specific survival proportions equally,
394 whereas “SSD weights” applies stable-stage-distribution weights from the annual *U* matrix.

Stage	Weighting	Years used	Median	Lower 95%	Upper 95%
Stage 1	No weights	4			
Stage 1	SSD weights	4			
Stage 2	No weights	12	0.235	-0.221	0.691
Stage 2	SSD weights	12	0.093	-0.670	0.856
Stage 3	No weights	12	0.238	-0.315	0.791
Stage 3	SSD weights	12	0.178	-0.353	0.708
Stage 4	No weights	12			
Stage 4	SSD weights	12			

395

396 Table S10. [Multisite robustness-analysis](#) stage-specific survival-temperature coefficients under
397 two point-estimate weighting scenarios across sites. The comparison shows how coefficient
398 estimates and interval widths change when no stage sample-size information is used and SSD
399 weights are substituted.

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Site	Stage	Weighting	Years used	Median	Lower 95%	Upper 95%
Haynes Creek	Stage 1	No weights	8			
Haynes Creek	Stage 1	SSD weights	8			
Haynes Creek	Stage 2	No weights	8	0.180	-0.401	0.761
Haynes Creek	Stage 2	SSD weights	8	-0.103	-0.748	0.542
Haynes Creek	Stage 3	No weights	8	0.232	-0.207	0.670
Haynes Creek	Stage 3	SSD weights	8	0.232	-0.369	0.833
Haynes Creek	Stage 4	No weights	7	-0.210	-0.625	0.206
Haynes Creek	Stage 4	SSD weights	7	-0.093	-0.745	0.559
Haynes Creek	Stage 5	No weights	8	0.028	-0.702	0.759
Haynes Creek	Stage 5	SSD weights	8	0.102	-0.372	0.577
McDevitt Creek	Stage 1	No weights	9	0.120	-0.485	0.725
McDevitt Creek	Stage 1	SSD weights	9	0.164	-1.030	1.350
McDevitt Creek	Stage 2	No weights	9	0.020	-0.601	0.642
McDevitt Creek	Stage 2	SSD weights	9	0.104	-0.881	1.090
McDevitt Creek	Stage 3	No weights	9	0.086	-0.605	0.777
McDevitt Creek	Stage 3	SSD weights	9	-0.152	-0.982	0.678
McDevitt Creek	Stage 4	No weights	9	-0.211	-0.608	0.186
McDevitt Creek	Stage 4	SSD weights	9	-0.152	-0.953	0.648
McDevitt Creek	Stage 5	No weights	9	0.283	-0.371	0.937
McDevitt Creek	Stage 5	SSD weights	9	0.248	-0.160	0.657
Sheep Corral Gulch	Stage 1	No weights	10			
Sheep Corral Gulch	Stage 1	SSD weights	10			
Sheep Corral Gulch	Stage 2	No weights	10	-0.317	-0.889	0.255

Site	Stage	Weighting	Years used	Median	Lower 95%	Upper 95%
Sheep Corral Gulch	Stage 2	SSD weights	10	-0.513	-0.938	-0.089
Sheep Corral Gulch	Stage 3	No weights	10	-0.460	-1.010	0.092
Sheep Corral Gulch	Stage 3	SSD weights	10	-0.247	-0.892	0.398
Sheep Corral Gulch	Stage 4	No weights	10	0.329	-0.877	1.530
Sheep Corral Gulch	Stage 4	SSD weights	10	0.520	-1.050	2.090
Sheep Corral Gulch	Stage 5	No weights	10	-1.010	-2.220	0.204
Sheep Corral Gulch	Stage 5	SSD weights	10	-0.098	-2.070	1.880

401

Table S11. Comparison of screened target studies that were included versus excluded under the current reproducible COMPADRE-based screen. Numeric entries are study-level means $\pm$ SD; life-form entries are numbers of studies with percentages in parentheses. Included studies have a higher mean proportion of boundary survival estimates, especially estimates of exactly 1.

Characteristic	Included	Excluded
Studies, n	23	69
Mean matrix dimension	5.92 $\pm$ 2.75	6.68 $\pm$ 5.05
Mean stage-specific survival	0.74 $\pm$ 0.14	0.76 $\pm$ 0.15
Proportion of survival estimates equal to 0	0.05 $\pm$ 0.10	0.03 $\pm$ 0.09
Proportion of survival estimates equal to 1	0.17 $\pm$ 0.17	0.08 $\pm$ 0.15
Proportion of survival estimates at the boundary (0 or 1)	0.23 $\pm$ 0.18	0.11 $\pm$ 0.17
Dominant life form: Herbaceous perennial	14 (60.9%)	40 (58.0%)
Dominant life form: Succulent	5 (21.7%)	8 (11.6%)
Dominant life form: Shrub	0 (0.0%)	4 (5.8%)
Dominant life form: Tree	2 (8.7%)	13 (18.8%)
Dominant life form: Palm	2 (8.7%)	4 (5.8%)

407 [Table S12. Distribution of boundary survival estimates and small stage-specific sample sizes](#)
 408 [across Analysis 1 population-level MPM entries, grouped by the COMPADRE life-form](#)
 409 [categories available for these studies and by broad matrix-dimension class. Percentages give the](#)
 410 [share of population entries with at least one boundary survival estimate or at least one stage with](#)
 411 [sample size below the stated threshold. Woody life-form categories are included for](#)
 412 [completeness but are sparsely represented in this subset.](#)

	Population entries, n	Any boundary survival estimate (%)	Any stage with N < 20 (%)	Any stage with N < 50 (%)	Mean minimum stage N
rennial	1	0.0	0.0	100.0	29.0
	53	45.3	34.0	58.5	66.3
	3	33.3	0.0	33.3	144.7
	1	100.0	0.0	0.0	190.0
	9	22.2	22.2	33.3	85.4
	1	100.0	100.0	100.0	8.0
	50	36.0	26.0	50.0	83.2
	18	61.1	44.4	66.7	43.6
413					

Table S13. Linear models relating displacement magnitude in Analysis 1 to boundary-survival predictors. Displacement is defined as the absolute difference between the point estimate and the median of the sampling distribution; life expectancy displacement is modelled on the log10 scale. Positive slopes indicate greater point-estimate displacement in matrices with more boundary survival structure.

Response	Predictor	Population entries, n	Slope	95% CI	P value	R ²
Shape displacement point - median	Number of boundary survival stages	68	0.010	[0.005, 0.016]	0.00037	0.176
Shape displacement point - median	Proportion of individuals in boundary-survival stages	64	0.093	[0.043, 0.143]	0.00045	0.181
Life expectancy displacement log10(point) - log10(median)	Number of boundary survival stages	68	0.062	[0.029, 0.095]	0.00035	0.177
Life expectancy displacement log10(point) - log10(median)	Proportion of individuals in boundary-survival stages	64	0.672	[0.370, 0.974]	< 1e-04	0.242

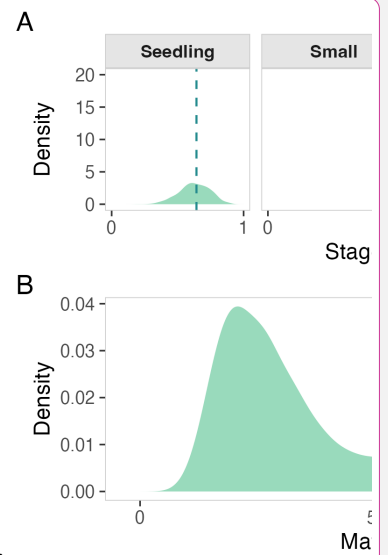
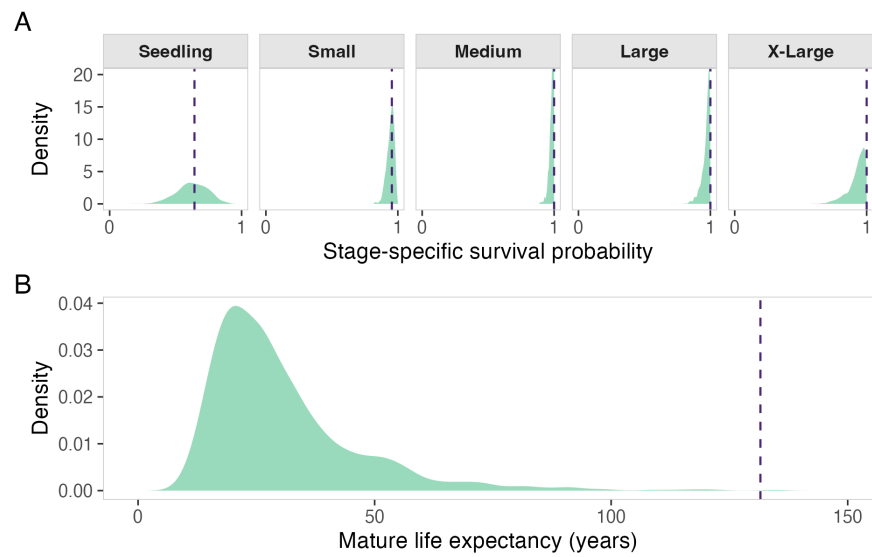
420 [Table S14. Sensitivity of Analysis 1 shape estimates to the quasi-stable-distribution truncation](#)
421 [threshold used when calculating the shape metric. Comparisons are against the 1% threshold](#)
422 [used in the main analysis, summarising the stability of shape estimates across the thresholds](#)
423 [tested; downstream regressions were not refitted.](#)

	Median IΔSI (point estimate)	95th percentile IΔSI (point estimate)	Max IΔSI (point estimate)	Median IΔSI (sampling median)	Sign changes in S	Me
	0.002	0.016	0.026	0.001	0	1
	0.004	0.050	0.064	0.003	2	-3

424

425 **Supplementary Figures**

426 **Figure S1.** Boundary-estimate diagnostic. Purple lines indicate point-estimate-derived quantities
427 and green densities indicate uncertainty-aware summaries from sampling distributions. When
428 stage-specific survival estimates are near 0 or 1, derived-parameter sampling distributions can be
429 displaced relative to point estimates, indicating potential inferential fragility.



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437 Figure S2. Illustration of survivorship trajectories under boundary survival transitions. Age/time
438 is shown in years. These cases can induce artefactual trajectory shapes and help explain
439 mismatches between point estimates and uncertainty-aware summaries.

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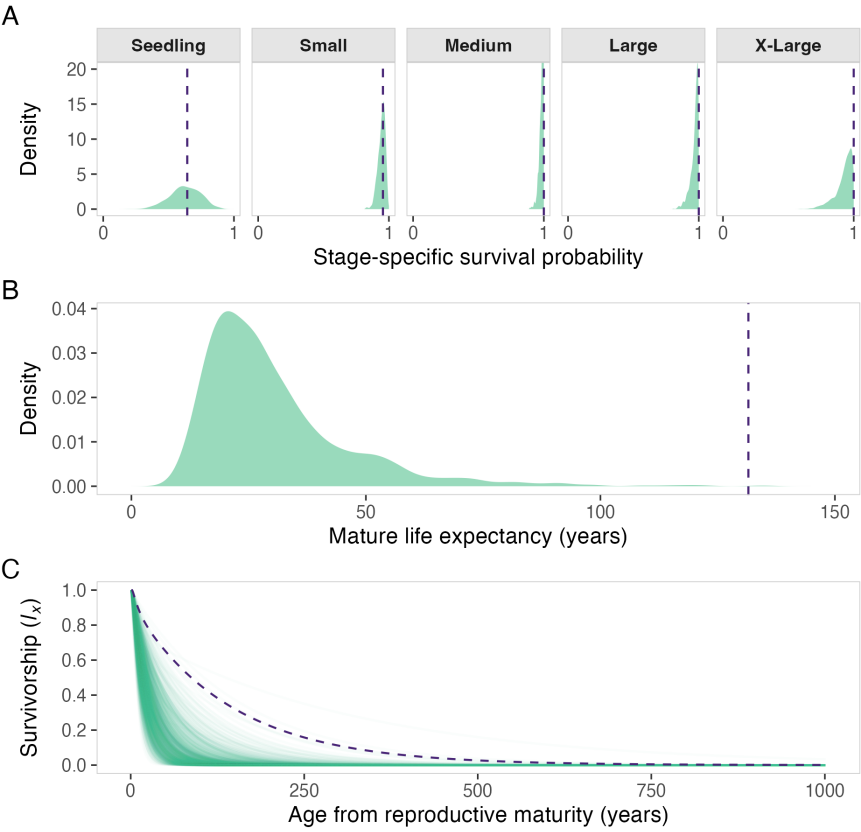


Figure S3. Displacement magnitude in Analysis 1 increases with boundary-survival structure. Each point is a population-level matrix. Panels compare the absolute difference between the point estimate and the sampling-distribution median against the number of boundary survival stages and the proportion of sampled individuals in those boundary stages, showing that matrices with more boundary-survival structure tend to have larger point-estimate displacement.

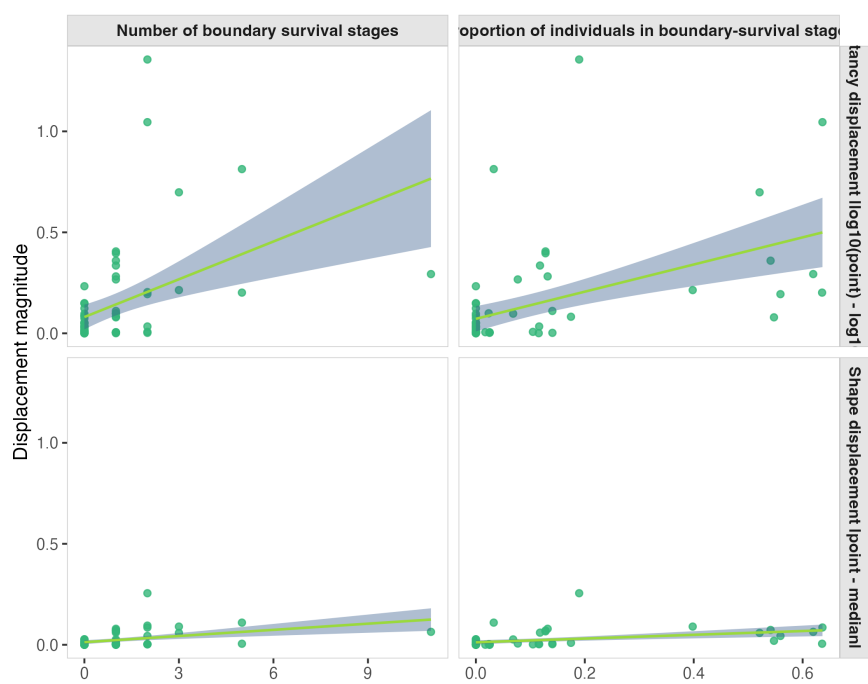


Figure S4. Sensitivity of Analysis 1 point estimates of mortality-trajectory shape to the quasi-stable-distribution truncation threshold. Point estimates calculated with 0.5% and 5% thresholds are plotted against the 1% threshold used in the main analysis, showing that most values lie close to the 1:1 line and that threshold choice makes little difference except for a small number of near-zero cases.

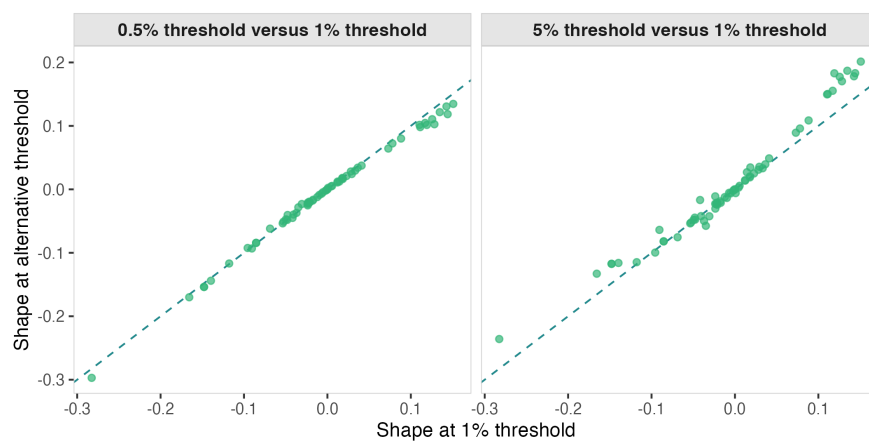


Figure S5. Multisite robustness analysis for *Astragalus scaphoides*. Columns show Haynes Creek (A, D, G), McDevitt Creek (B, E, H), and Sheep Corral Gulch (C, F, I). Rows show monthly-lag temperature coefficients (A-C), recruitment-temperature fits (D-F), and stage-specific survival-temperature coefficients (G-I) for point-estimate (purple) and uncertainty-aware (green) formulations. Recruitment is seedlings per reproductive plant, and stage-specific survival coefficients are on the logit scale (β). Survival intervals are 95% Wald confidence intervals; non-estimable stages are labelled rather than plotted. Differences between survival models reflect weighting and dispersion assumptions as well as the use of sample sizes. Recruitment results are closely aligned between formulations within sites.

